Analytical Chemistry Cumulative Exam
April 17, 2009
Basic Luminescence Concepts

1. (6 points) Define fluorescence. How does a molecule fluoresce?

2. (10 points) What is quantum yield? Write an equation for quantum yield for fluorescence in terms of the relative rate constants for the deactivation processes. Be sure to name or label all of your variables.

3. (10 points) Name three deactivation processes (not including fluorescence) and what they are due to. Draw a Jablonski diagram to help you in your explanations.

4. (12 points) In the most general terms, define and compare electronic singlet and electronic triplet states. What kinds of luminescence can result from the transition of an electron in an excited singlet state to a ground singlet state and from an excited triplet state to a ground singlet state.

5. (6 points) Define phosphorescence. How does a molecule phosphoresce?

6. (6 points) Describe chemiluminescence. Give one example of chemiluminescence in nature or in a commercial product.

7. (20 points) Draw an instrumentation block diagram that shows the general instrumental components that are needed to obtain a fluorescence spectrum. Label the drawing and components. Suggest a specific kind of component that could be used for each "block" in the diagram. (For example, if you need a light source, you need to suggest a specific kind of lamp, too.)

8. (10 points) Describe one procedure of how fluorophores can be used to quantify a specific non-fluorescing analyte in a sample mixture.

9. (20 points) For a given fluorescing species in a solution, how would the following changes likely affect the fluorescence intensity (increase the fluorescence, decrease the fluorescence, or leave the fluorescence unchanged)? Explain your answer to each.

(a) increase the viscosity of the solution
(b) increase the temperature
(c) add a halogenated solvent
(d) increase the oxygen content
(e) increase the concentration of the fluorescing species so that it is highly concentrated